

Research Question

PDE4445

2025 – 2026

Outline



- What is “research”?
- The research process
- Finding a research topic
- Developing a research question

What *really* is research?



***“Original investigation
undertaken in order to
gain knowledge and
understanding.”***

- Higher Education Funding Council for England

What *really* is research? 🧐

- Process of systematic enquiry – Asking “what”, “why” and “how”
- Learning something new, something you didn’t know already

Why do it? 💪

- Explaining observed phenomena, processes or patterns
- New theories or modifications of existing theories
- Improving processes, analysis, effectiveness of play, patterns, workflows

The Research Process



The Research Process

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1. Select Topic
2. Define Research Question
3. Create Title
4. Conduct Subject Area Review [i.e. literature review]
5. Design Research Approach & Data Collection Methods
6. Write Research Proposal
7. Collect Data
8. Analyse Data
9. Present Findings
10. Discuss Findings and make Conclusions
11. Make Recommendations

The Research Question ?

The Research Question ?

The key to successful research is not finding the answers, but finding the right questions to ask

A research question is –

“A clear, concise statement of a problem or issue which you are investigating and attempting to find an answer to.”

Research questions should be ?

- Feasible, specific, non-trivial
- Must address a real and genuine problem (not a hypothetical or “toy” question, or a non-issue)
- A recognised issue or question
- Sufficiently complex, where the answers are not obvious or known already
- If you know the answer to your research question already, or it has a simple yes/no answer, you need to revise your question

The Research Question ?

Fundamental decision for the rest of the module

Your research question will determine your research methods and your choice of data collection techniques

Your research question will guide your project as a whole